



Hypercalcemic Ovarian Carcinoma (Small Cell Carcinoma of the Ovary, Hypercalcemic Type (SCCOHT)): A Case Series and Review of Literature of a Rare Malignancy

Chandan C S¹ · Abdul Waheed Mir² · Ab Wahid Mir¹ · Firdous Ahmad Dar¹ · Syed Besina Yasin³

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Abstract

Small cell carcinoma of the ovary, hypercalcemic type (SCCOHT) is one of the rare and aggressive cancers occurring mainly in adolescents and young women. It accounts for less than 0.01% of all ovarian malignancies, with around 500 cases reported to date in the medical literature. It has been postulated that these cases could be due to genetic alterations, and in 2014, studies discovered that SCCOHT is characterized by both germline and somatic deleterious mutations in SMARCA4. In this case series, we report two cases diagnosed with carcinoma ovary and found to be having mutations in SMARCA4. The patients in this case series were relatively young and both had germline SMARCA4 mutation. The first patient had FIGO stage IV disease and the second patient had FIGO stage III disease. The first patient started with paclitaxel plus carboplatin and the second patient started with gemcitabine plus docetaxel. One patient succumbed to disease within 1 year and another patient is alive at present with progression of disease. SCCOHT is very aggressive and needs diagnosis and treatment at an early stage. Early diagnosis and proper treatment may prolong survival. There is a need for evaluation of the possible role of targeted systemic therapeutic options as the conventional regimens are rarely sufficient.

Keywords Ovarian carcinoma · SCCOHT · SMARCA4 · Hypercalcemia

Introduction

Small cell carcinoma of the ovary, hypercalcemic type (SCCOHT) is one of the rare and aggressive cancers occurring mainly in adolescents and young women. It accounts for less than 0.01% of all ovarian malignancies [1], with around 500 cases reported to date in the medical literature. In more than half of the cases, this cancer is associated with hypercalcemia and exact pathophysiology of hypercalcemia is unknown; however, in some cases, it may be related to secretion of PTH-RP (parathyroid hormone-related

protein) [2]. It has been postulated that these cases could be due to genetic alterations, and in 2014, studies discovered that SCCOHT is characterized by both germline and somatic deleterious mutations in SMARCA4. It was shown that SMARCA4 appears as the only recurrently mutated gene in SCCOHT [3]. Because of the rarity of the disease, there are no evidence-based treatment recommendations and these cases have poor prognosis even when diagnosed at an early stage.

In this case series, we report two cases diagnosed with carcinoma ovary and found to be having mutations in SMARCA4.

Case 1

A 30-year-old female patient with no underlying comorbidities presented with dull aching pain abdomen.

The patient was evaluated for the same. Hemogram was unremarkable but biochemistry showed serum calcium of 12.09 g/dl. The patient had undergone exploratory laparotomy at an outside hospital, where the patient

✉ Chandan C S
sinchan294930@gmail.com

¹ Department of Surgical Oncology, Sher-I-Kashmir Institute of Medical Sciences, Srinagar, J&K, India

² Department of Critical Care Medicine, Sher-I-Kashmir Institute of Medical Sciences, Srinagar, J&K, India

³ Department of Pathology, Sher-I-Kashmir Institute of Medical Sciences, Srinagar, J&K, India

was labeled as unresectable and the biopsy of the adnexal mass was taken. The patient reported to our outpatient department. On examination, the patient had an infra umbilical laparotomy incision with sutures in situ with a large 3×2 cm left supra clavicular lymph node. Biopsy of the left supraclavicular lymph node was taken which showed histological features suggestive of metastatic deposits from high-grade carcinoma ovary. Contrast enhanced CT (CECT) (chest/abdomen/pelvis) showed pelvic mass. Tumor markers were within normal limits. GI screening was unremarkable. IHC of both primary and neck node biopsy showed tumor cells showing loss of BRG-1 expression and INI-1 was retained. The other causes of hypercalcemia were ruled out. The patient was diagnosed with SCCOHT and SMARCA4-deficient and offered chemotherapy (paclitaxel and carboplatin), but disease continued to progress and the patient succumbed to the disease within 1 year.

Case 2

A 40-year-old unmarried female patient with no underlying comorbidities had presented with complaints of pain in the abdomen and irregular cycles.

USG (A/P) showed a bulky uterus with large multiple heterogeneous lesions in both walls of the uterus measuring 15×11.1 cm suggestive of fibroid and hyperplastic endometrium (thickness 30 mm). MRI pelvis shows diffuse uterocervical altered signal mass with extraserosal extensions with diffusion restriction, thick uterine endometrium with multiple small submucosal leiomyomas, bilateral ovarian and POD deposits, and bilateral iliac and para-aortic LAP. CECT (C/A/P) features were suggestive of locally advanced uterine sarcoma with bilateral ovarian deposits, multiple enlarged iliac and retroperitoneal nodes with mild to moderate ascites, and pouch of douglas (POD) deposits. Biochemistry showed serum calcium 9.9 g/dl. Serum CEA — 1.42, CA 19.9 — 8.5, CA125 — 788, AFP — 2.83, beta HCG — 0.25, and LDH — 353. The patient underwent laparotomy in an outside hospital but was labeled inoperable. The patient was again taken for exploratory laparotomy in our hospital. Intra-operative findings revealed 15×15 cm mass replacing uterus, complex cyst of bilateral ovaries, pelvic peritoneal, omental and diaphragmatic deposits with mild ascites. Omental and pelvic peritoneal biopsies were taken. HPE showed undifferentiated high-grade malignant ovarian tumor of SMARCA4-deficient type (Figs. 1 and 2). The tumor cells express CD99 diffusely and show complete loss of BRG-1 expression (Figs. 3 and 4). Tumor cells did not express CK, EMA, PAX-8, Cyclin-D1, CD10, PR, SMA, desmin, S-100 protein, CD34,

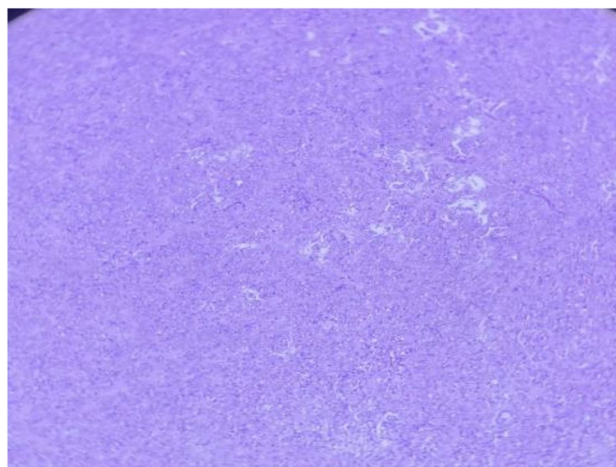


Fig. 1 H&E photomicrograph showing small round cells arranged in sheets with focal rosette formation (10 $\times$)

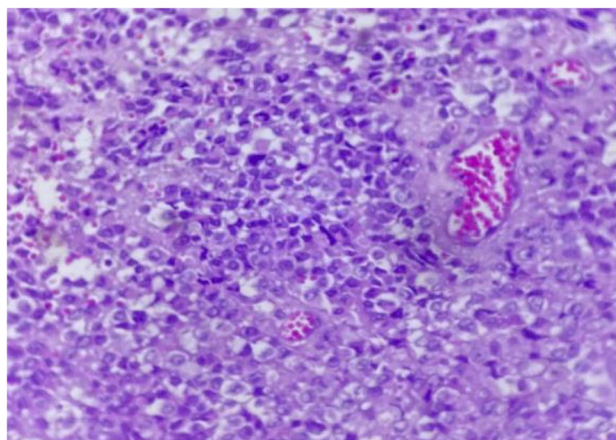


Fig. 2 H&E photomicrograph showing small round cells with high N/C ratio hyperchromatic nucleus and scant cytoplasm in sheets with perivascular rosetting

LCA, synaptophysin, WT-1, TLE-1, NKX2.2, or BCOR. The INI-1 expression is retained. The patient was started chemotherapy (gemcitabine + docetaxel). The patient is alive with progression of disease.

Discussion and Review of Literature

Small cell carcinoma of the ovary is a rare malignant tumor. The clinical and pathologic aspects of this malignancy were first described by Scully in 1979 [4]. In his description, he noted that these neoplasms had the characteristic morphologic appearance of small hyperchromatic

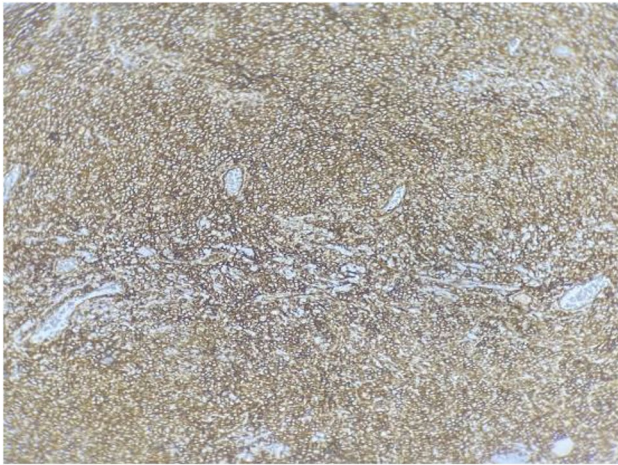


Fig. 3 IHC photomicrograph showing diffuse strong positivity for CD99

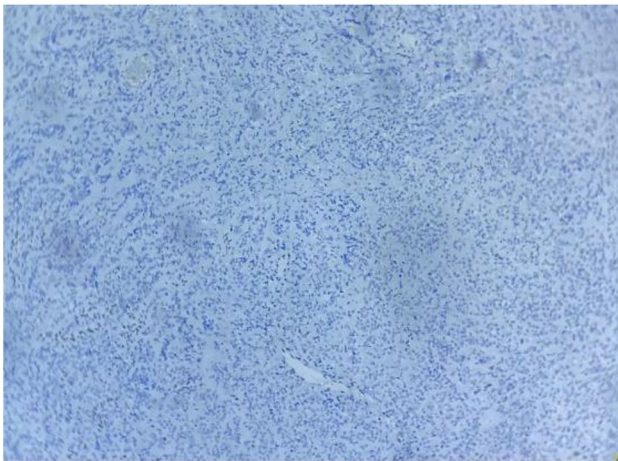


Fig. 4 IHC photomicrograph showing loss of BRG-1 expression

cells with scant cytoplasm and brisk mitotic activity. The occurrence was more in young females, and patients had typically hypercalcemia. SCCOHT is currently classified as a miscellaneous neoplasm in the 2014 World Health Organization (WHO) Classification of Tumors of Female Reproductive Organs [5].

SCCOHT is predominantly or exclusively composed of small round cells with scant cytoplasm. The discovery of *SMARCA4* mutations in almost all SCCOHT tumors resulted in the development of a *SMARCA4* (also known as BRG1) antibody, which is highly useful in the diagnosis of this neoplasm and distinguishing from mimicking tumors. One of the original publications describing germline and somatic *SMARCA4* mutations in these neoplasms showed loss of *SMARCA4* nuclear immunoreactivity in 51 of 54 (94%) cases. Subsequent studies showed that over 95% of these

neoplasms exhibit loss of nuclear immunoreactivity with this marker, making it an important diagnostic tool, although not all cases are negative [6].

Familial SCCOHT is characterized by an autosomal dominant transmission and could be due to underlying mutations in a single gene or a family of genes. Mutations in familial diseases might be causing molecular events in sporadic cases. Hence, efforts have been taken to describe comprehensive molecular characterization in SCCOHT. *Tp53* mutations were occasionally reported in SCCOHT [7], but *TP53* mutations, the hallmark of high-grade serous carcinoma, were extremely rare in SCCOHT. The other mutations like *BRCA1*- and *BRCA2*- mutations and *KRAS* or *BRAF* mutations were checked in SCCOHT. *BRCA1*- and *BRCA2*-germline mutation carriers have been highly implicated in developing ovarian carcinomas, predominantly high-grade serous carcinoma and rarely other histotypes, such as non-small neuroendocrine carcinoma [8]. However, to date, no SCCOHT case has been found to be associated with *BRCA1*- or *BRCA2*-germline mutations. Also, SCCOHT is not associated with mutations in classical oncogenes including *KRAS* or *BRAF*. Hence, it becomes clear that SCCOHT differs from ovarian carcinoma by displaying a low mutational burden and little chromosomal abnormalities. However, in spite of being low mutation load cancer, SCCOHT displayed an immunogenic microenvironment resembling the tumors with immune checkpoint blockade, characterized by a high programmed death ligand-1 (PD-L1) expression, strongly associated with T-cell infiltration, and increased expression genes related to Th1 and cytotoxic cell function [9]. The inconsistencies between mutation burden and immunogenicity underscored the requirement for additional biomarkers for tumor recognition in SCCOHT.

The age at diagnosis has ranged from 7 months to 56 years, with an average age of 23.9 years. In one study, 26 of 60 patients (43%) had a germline *SMARCA4* mutation, including all patients diagnosed before the age of 15 years. Women with germline mutations present at a significantly younger age than those without mutations [10]. The patients in our case series were relatively young and both had germline *SMARCA4* mutation consistent with case reports and systematic reviews published in literature.

Due to the lack of prospective studies, recommendations for management are based on small case series and the strategies remain heterogeneous. Despite the absence of data, the general principles of primary cytoreductive surgery for epithelial ovarian cancer apply to patients with SCCOHT, with the goal of complete surgical resection leaving no gross disease. A multimodal approach including radical surgery, chemotherapy, and radiotherapy after discussion at a multidisciplinary tumor board is essential [11]. Fertility-sparing surgery may be considered in comprehensively surgically

staged stage IA patients without germline *SMARCA4* mutations if they desire future pregnancies, although radical surgery for all patients remains the norm. Adjuvant chemotherapy is recommended for all stages, generally cisplatin- and etoposide-based combination regimens (e.g., BEP: bleomycin, etoposide, and cisplatin; VPCBAE: vinblastine, cisplatin, cyclophosphamide, bleomycin, doxorubicin, and etoposide; and PAVEP: cisplatin, doxorubicin, and etoposide cyclophosphamide). For patients in whom initial surgery is not feasible (e.g., stage IV disease, unresectable disease, and medically unfit patient), administration of chemotherapy and interval cytoreductive surgery may be considered on an individual basis. High-dose chemotherapy (HDC) with autologous stem cell transplantation rescue following a complete response to initial chemotherapy, with or without surgery, may also be considered [12].

In our case series, the first patient had FIGO stage IV disease and the second patient had FIGO stage III disease. The first patient was started with paclitaxel plus carboplatin and the second patient started with gemcitabine plus docetaxel as there was a lack of clear protocol for systematic treatment in literature. One patient succumbed to disease within 1 year and another patient is alive at present with progression of disease, again consistent with poor prognosis associated with these aggressive tumors.

Conclusion

SCCOHT is a rare disease, but in the present era of increased use of molecular and genetic testing, more and more cases will be diagnosed. The disease is very aggressive and needs diagnosis and treatment at an early stage. The young patients presenting with ovarian malignancy may be evaluated for the disease with small cell morphology. Early diagnosis and proper treatment may prolong survival. Also, there is a need for evaluation of the possible role of targeted systemic therapeutic options as the conventional regimens are rarely sufficient.

Declarations

Informed Consent Written informed consent has been obtained from the patient(s) allowing the use of her case information for clinical research.

Conflict of Interest The authors declare no competing interests.

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